

	Pimpri Chinchwad Education Trust's Pimpri Chinchwad College of Engineering	
ASSIGNMENT – 5		

Integrate Jenkins with a Git repository . Configure Jenkins to pull code from GitHub

Objective:

To perform continuous integration by pulling code from a GitHub repository using Jenkin **Theory:**

Jenkins–GitHub Integration:

Jenkins-GitHub integration enables Jenkin's to fetch the source code automatically from GitHub and trigger builds whenever changes related to commits are made in the repository. This helps in achieving Continuous Integration where, after every commit or pull request, the code would be built and tested immediately. Jenkins integrates with GitHub using plugins and webhooks that trigger builds automatically after events like code commits or pull requests to provide fast feedback and early error detection. The solution enhances collaboration among different developers and guarantees that the latest code has been tested and prepared for deployment.

Procedure:

Sample Template

Step 1: Install Jenkins

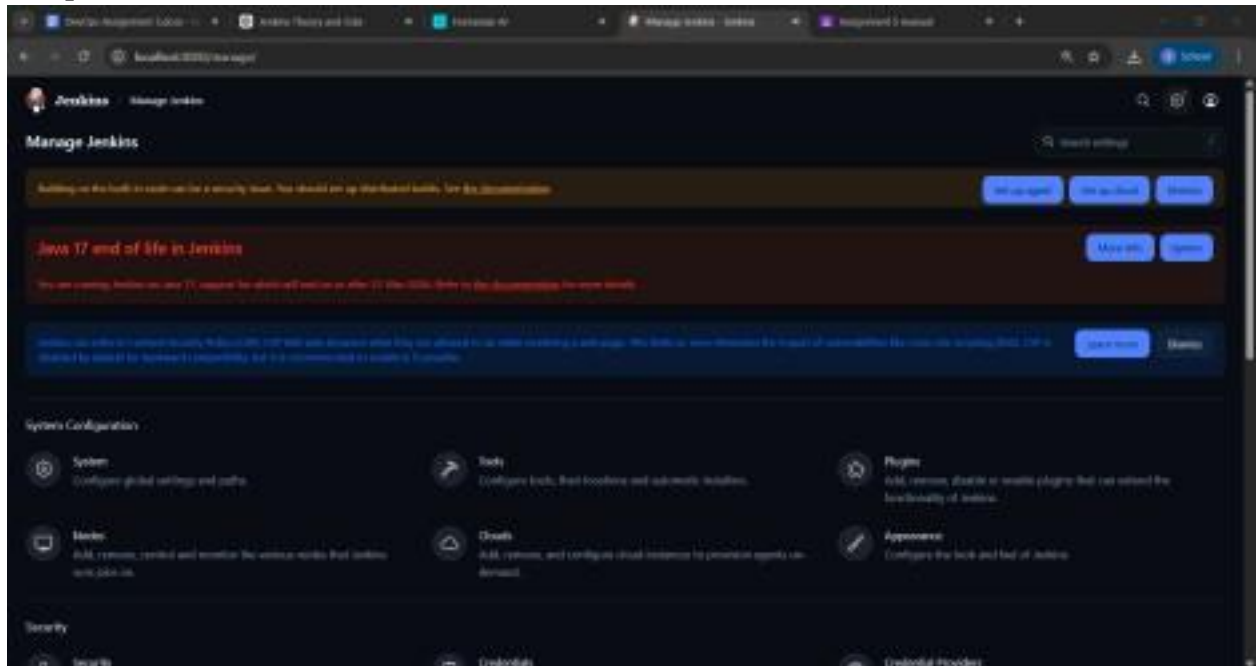
Steps to Install Jenkins (Simple – Windows)

1. Download Jenkins from the official Jenkins website.
2. Make sure Java (JDK) is installed on your system.
3. Run the downloaded Jenkins installer (.msi file).
4. Follow the setup wizard and choose the installation path.
5. During installation, Jenkins will be set up as a Windows service.
6. After installation, open a browser and go to <http://localhost:8080>

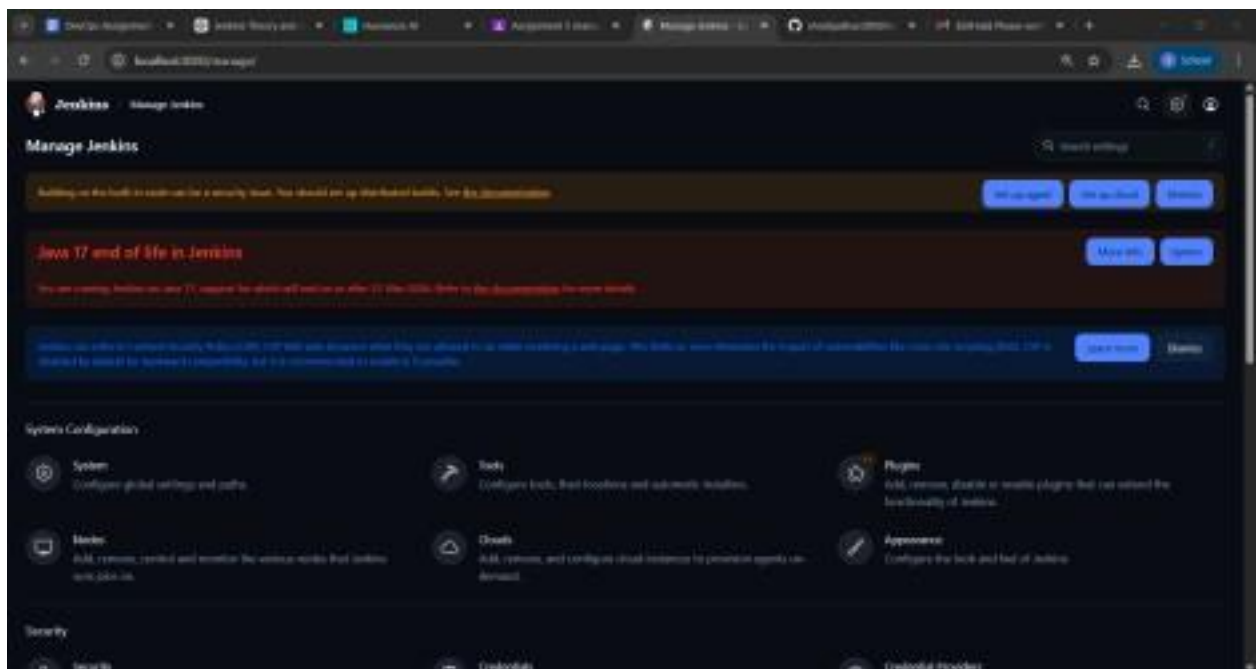
7. Enter the initial admin password (shown on screen or found in the Jenkins folder).
8. Click Install suggested plugins.
9. Create an admin user and finish the setup.

Step 2: Install Required Plugins

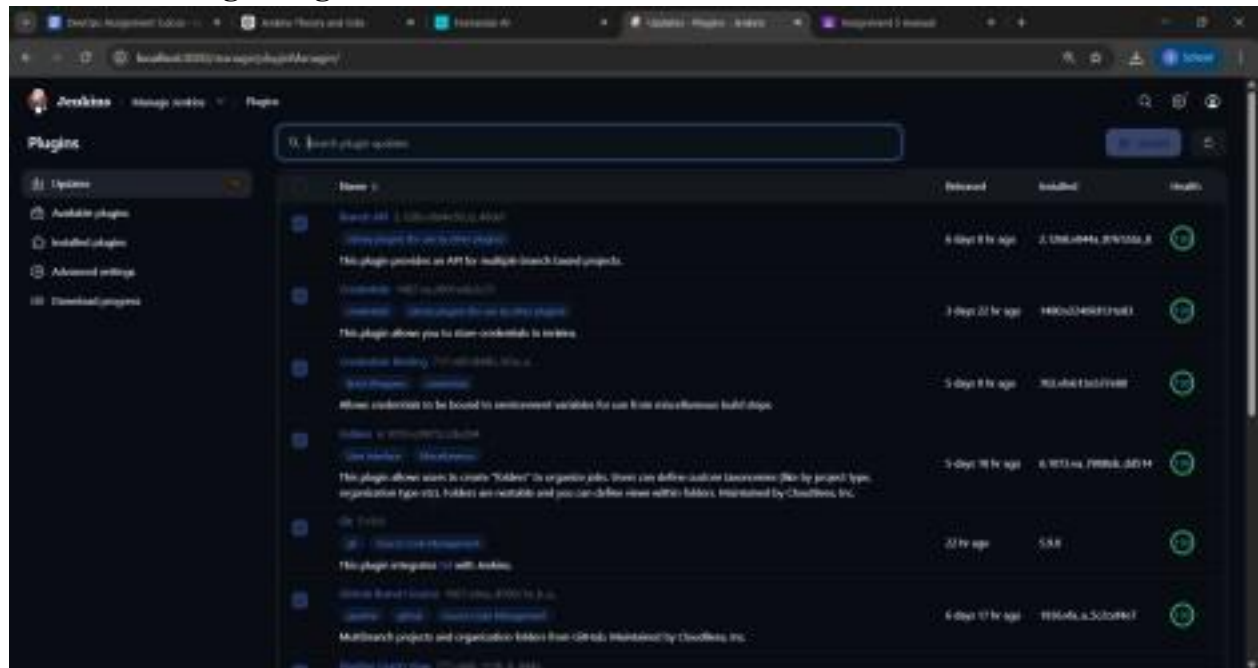
1. Open Jenkins Dashboard.



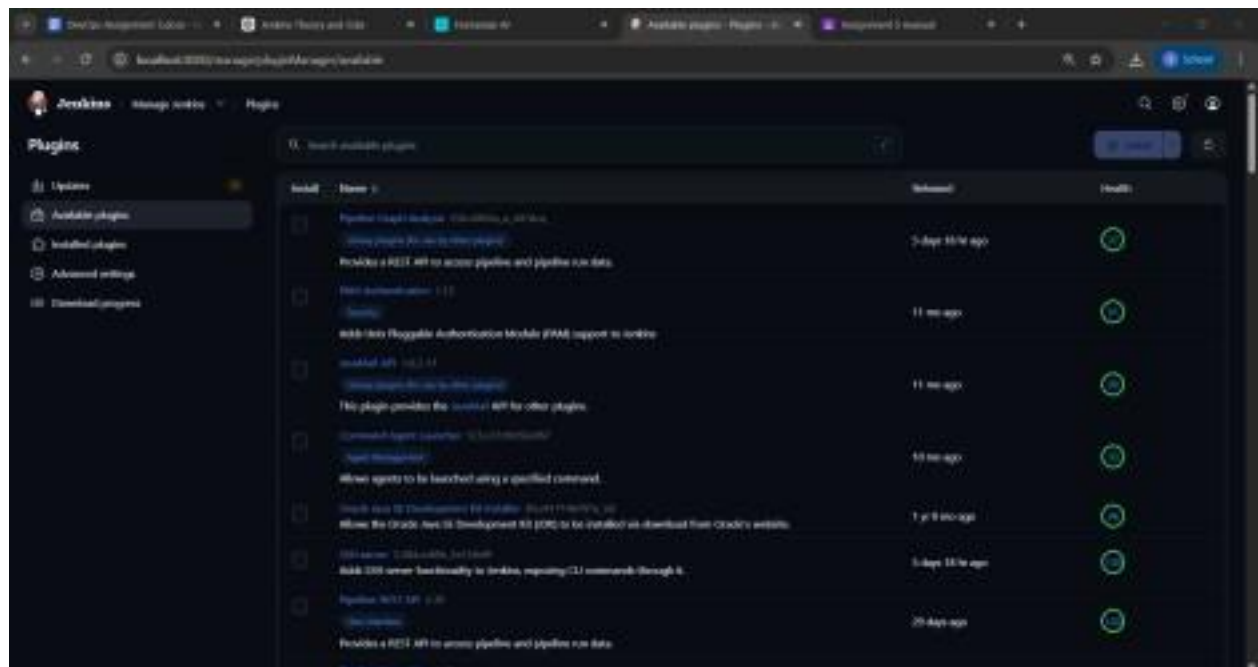
2. Click on Manage Jenkins.



3. Select Manage Plugins.



4. Open the Available tab.

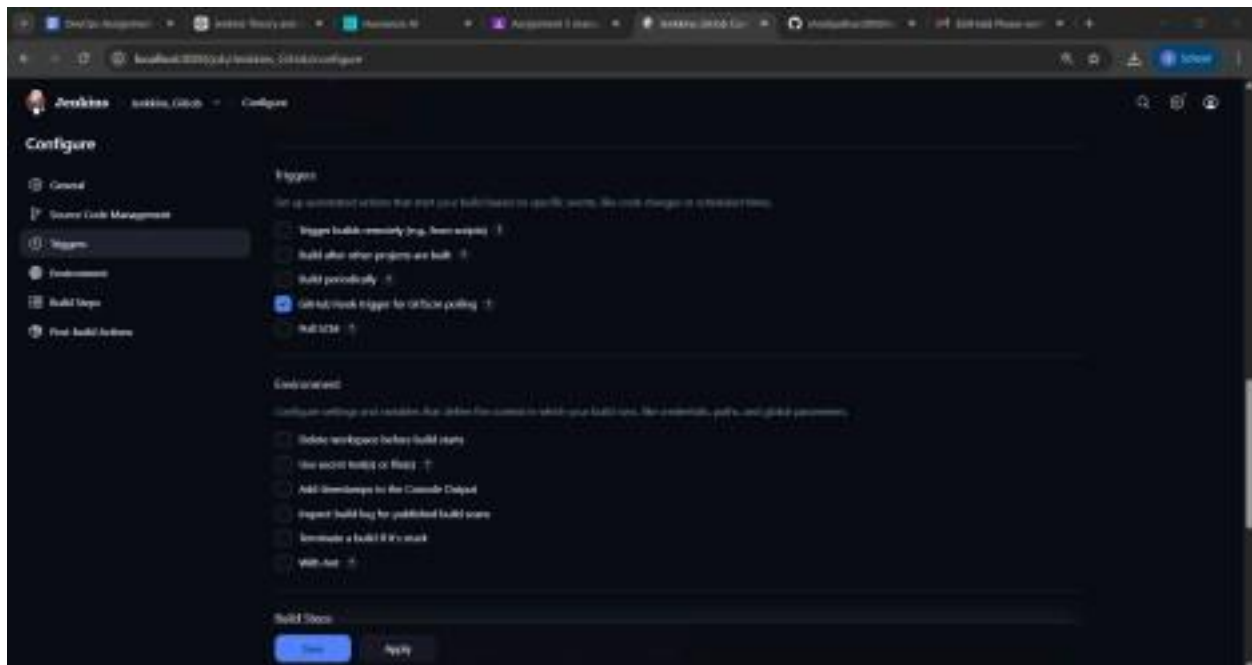
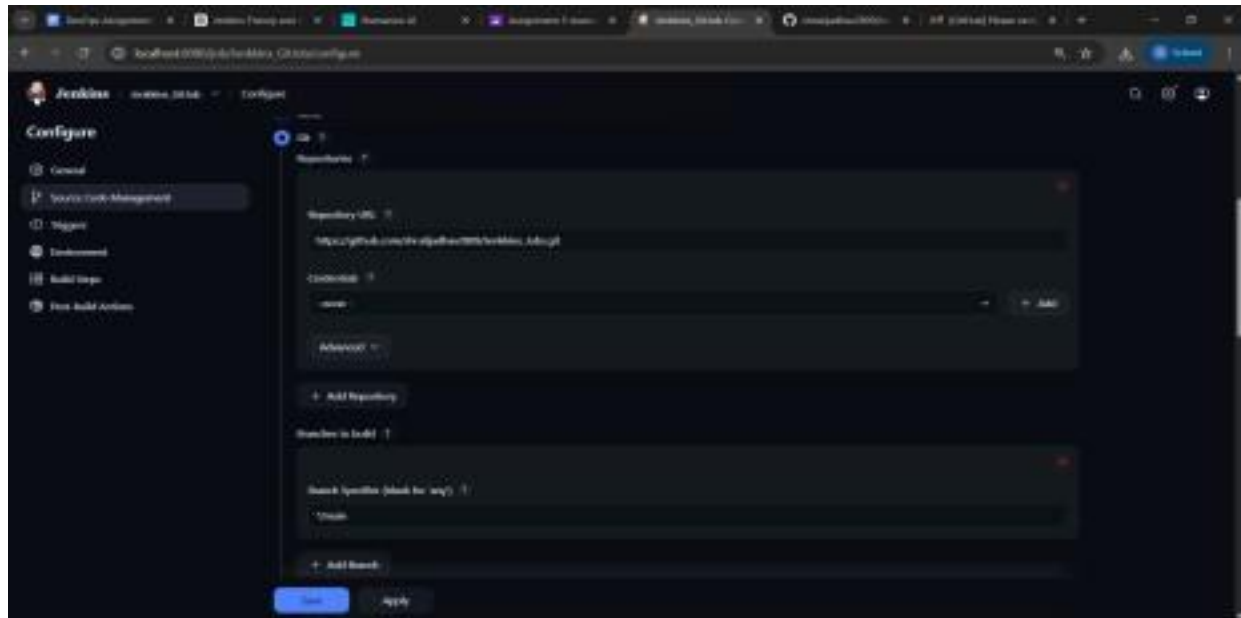


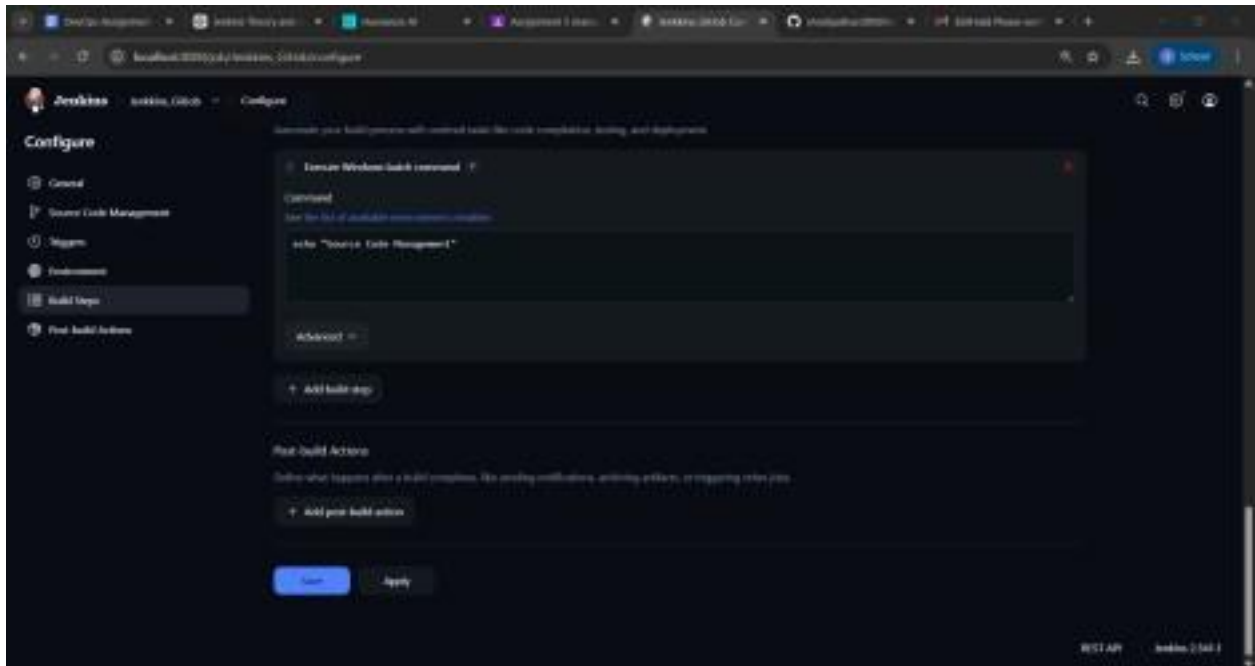
5. Search for the required plugin.

6. Select the plugin.

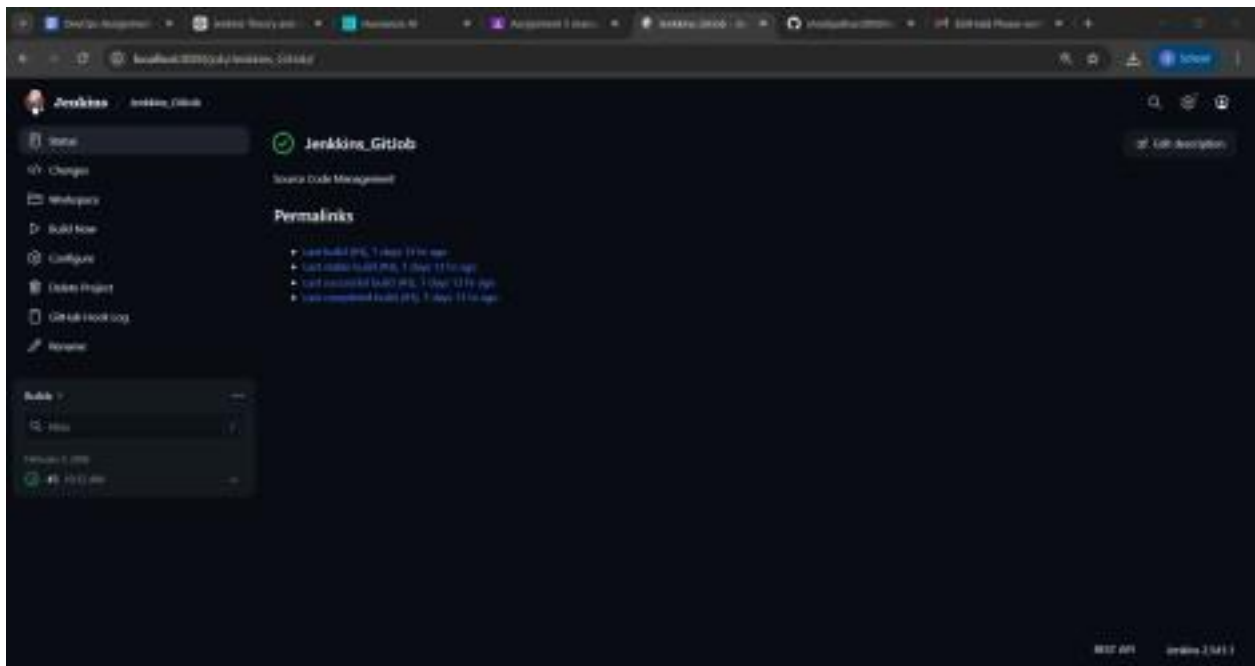
7. Click **Install without restart** (or **Install after restart**).

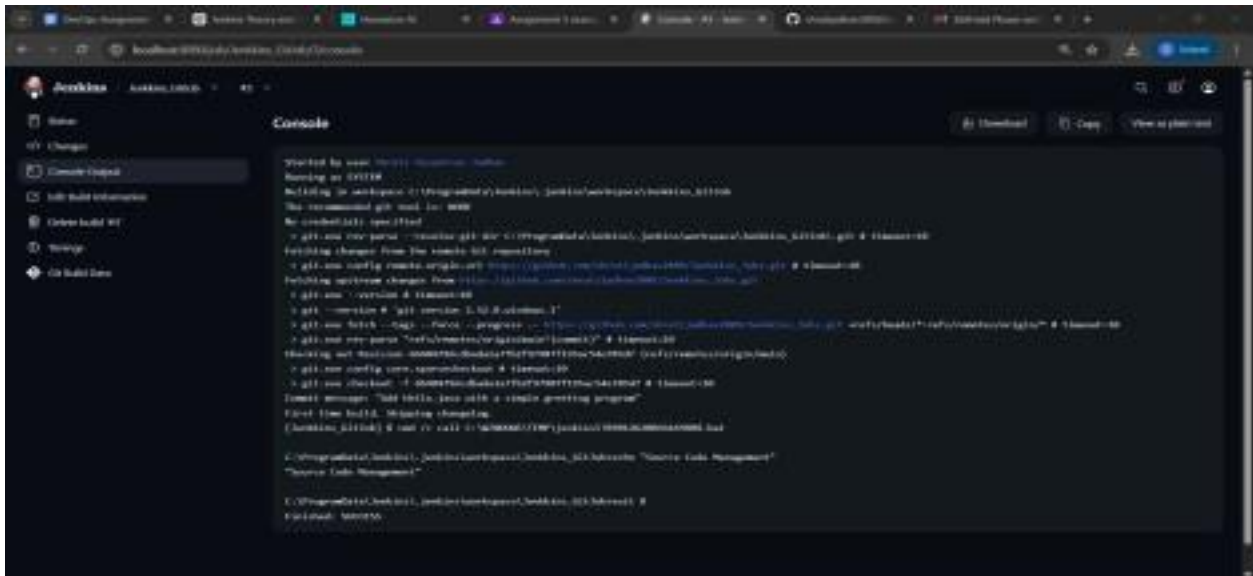
Step 6: Configure Source Code Management





Step 7: Build the Job and Display output





FAQ:

1. How does Jenkins integrate with GitHub?

Jenkins integrates with GitHub by connecting to GitHub repositories using plugins and webhooks. Jenkins pulls source code from GitHub using Git, and GitHub webhooks automatically trigger Jenkins jobs whenever code is pushed or a pull request is created.

This enables Continuous Integration by ensuring that every code change is automatically built and tested.

2. Which plugin is required to integrate GitHub with Jenkins?

The GitHub Plugin is required to integrate GitHub with Jenkins. Additionally, the Git Plugin is used to fetch source code from GitHub repositories and manage Git operations within Jenkins.

Conclusion :

By installing Jenkins, configuring Git, integrating GitHub, and using the required plugins, Jenkins enables automated build and integration of code changes. This setup supports Continuous Integration, reduces manual effort, and ensures faster and more reliable software development.